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4811

Tutorial 1 - Algebra - MAP 2301

① (I) $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$

Solution: we want show $A \cap (B \Delta C) \subseteq (A \cap B) \Delta (A \cap C)$ and $(A \cap B) \Delta (A \cap C) \subseteq A \cap (B \Delta C)$

* $A \cap (B \Delta C) \subseteq (A \cap B) \Delta (A \cap C)$

let $x \in A \cap (B \Delta C)$; x be an arbitary element
$x \in A$ and $x \in ((B-C) \cup (C-B))$ ($\because$ By defintion)
$x \in A$ and $x \in B-C$ or $x \in C-B$

Case 1: $x \in (B-C)$
$x \in B$ But $x \notin C$, already $x \in A$ So $x \in A \cap B$ $\{x \notin A \cap C\}$

Case 2: $x \in (C-B)$
$x \in C$ But $x \notin B$, alredy $x \in A$ so $x \in A \cap C$ $\{x \notin A \cap B\}$

Now we can say $x \in (A \cap B) \Delta (A \cap C)$ $\{\because (A \cap B)-(A \cap C) \cup (A \cap C)-(A \cap B)\}$
Hence; $A \cap (B \Delta C) \subseteq (A \cap B) \Delta (A \cap C) \longrightarrow$ ①

* $(A \cap B) \Delta (A \cap C) \subseteq A \cap (B \Delta C)$
let $y \in (A \cap B) \Delta (A \cap C)$; y be an arbitary element
$y \in (A \cap B) - (A \cap C)$ or $(A \cap C) - (A \cap B)$

Case 1:
$y \in (A \cap B) - (A \cap C)$
So $y \in A$, $y \in B$, $y \notin C$
$\Rightarrow y \in A \cap B$ and $y \in (B-C)$
$\Rightarrow y \in A \cap (B-C)$

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Cas 2: $y \in (A \cap C) - (A \cap B)$

so $y \in A$ $y \in C$ and $y \notin B$

$\Rightarrow y \in (A \cap C)$ and $y \notin B$

$\Rightarrow y \in (A \cap C)$ and $y \in (C - B)$

$\Rightarrow y \in A \cap (C - B)$

using case 1, 2 $\Rightarrow$ we can say $y \in A \cap (B \Delta C)$

Henc: $(A \cap B) \Delta (A \cap C) \subseteq A \cap (B \Delta C) \rightarrow$ (2)

(1), (2) $\Rightarrow$

$A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$ //

II. $A \Delta B = \emptyset$, then $A = B$

Assume $A \Delta B = \emptyset$ and $A \neq B$ $\{x \in A, y \in B\}$

Case 1: $x \in A$ and $x \notin B$

Since x is in A and not in B

$\Rightarrow x \in (A - B)$

But, since $A \Delta B = \emptyset$ this implies $x \in A - B$ and $x \in B - A$, which contradicts the assumption

So $A \Delta B = \emptyset$

Case 2: $y \in B$ and $y \notin A$

Similarly, since y is in B and not in A

$\Rightarrow y \in (B - A)$

Since $A \Delta B = \emptyset$ this implies $y \in A - B$ and $y \in B - A$, which contradicts the assumption

So $A \Delta B = \emptyset$

So our $A \neq B$ assumption is wrong

Hence $A = B$

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Not Only that $A = B$ we can say $A \Delta B = \emptyset$ (reverse the prove)

$\therefore A \Delta B = (A - B) \cup (B - A)$

$A = B$

$= (A - A) \cup (A - A)$

$= \emptyset \cup \emptyset$

$\therefore A \Delta B = \emptyset$ //

(2) (i) $\therefore A \cup B = A + B - A \cap B$

$= 70 + 45 - 25$

$= 90$

(ii) 70 Student offerd chemistry

45. Students offerd chem and maths.

So; $70 - 45$

25 chem only.

So; phy only + chem only $= 45$.

phy only + $25 = 45$

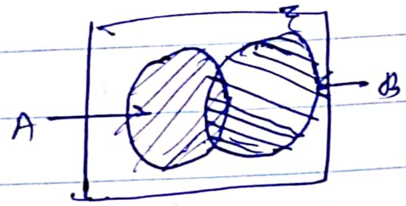
phy only $= 20$

(iii) No; of stundent not offered maths; $100 - 45$.

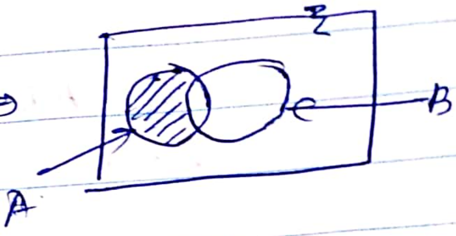
55

3. (a) $A - B = A \cap B'$

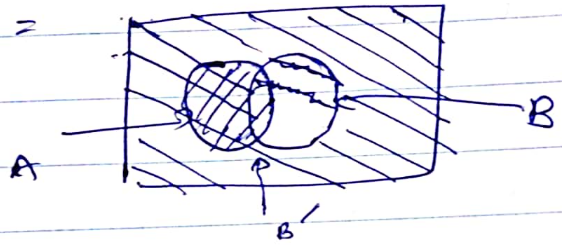
(I) LHS =



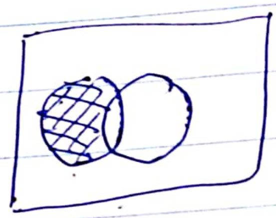
$A - B \Rightarrow$



RHS =

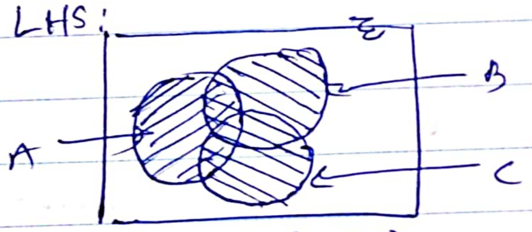


$A \cap B' \Rightarrow$



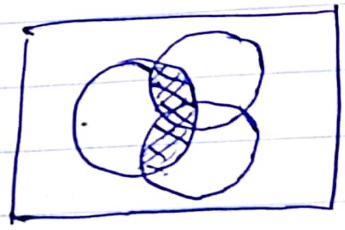
LHS = RHS //

(II) LHS;



$A \cap (B \cup C)$

$\Rightarrow$



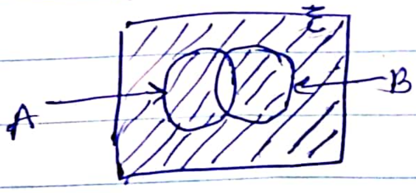
RHS;

$(A \cap B) \cup (A \cap C)$

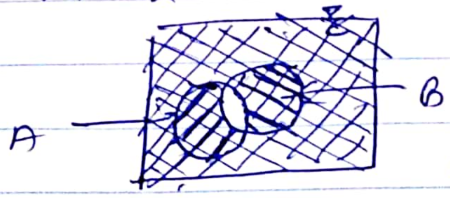
LHS = RHS

$(A \cap B) \cup (A \cap C) = A \cap (B \cup C)$ //

(III) LHS: $(A \cap B)'$

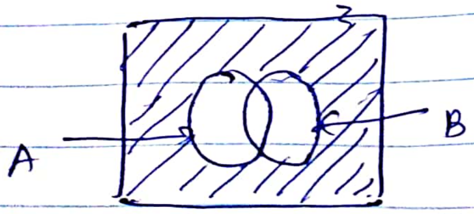


RHS: $A' \cup B'$

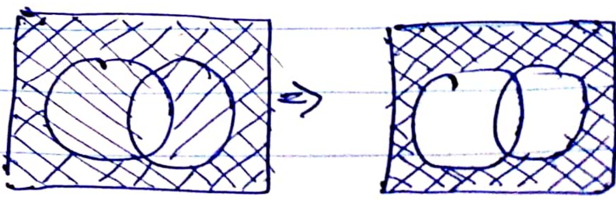


LHS = RHS

(IV) LHS: $(A \cup B)'$

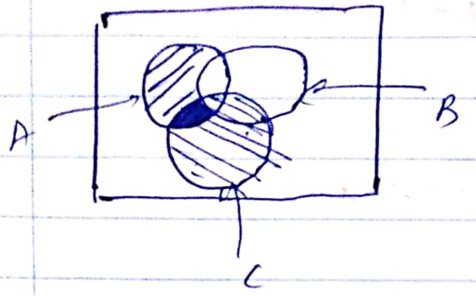


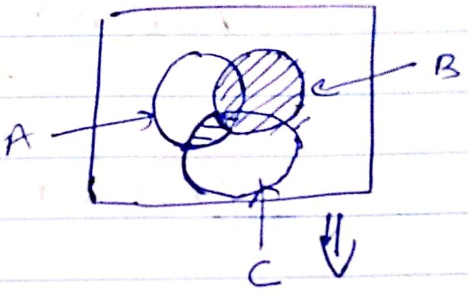
RHS: $A' \cap B'$

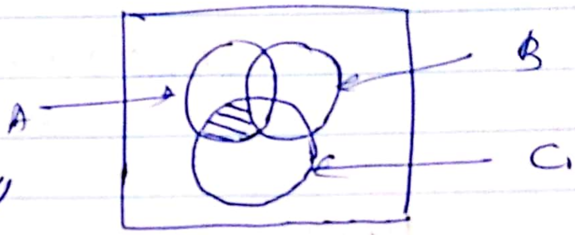


LHS = RHS

(v) LHS: $(A-B)\cap C$ RHS: $(A\cap C)-B$



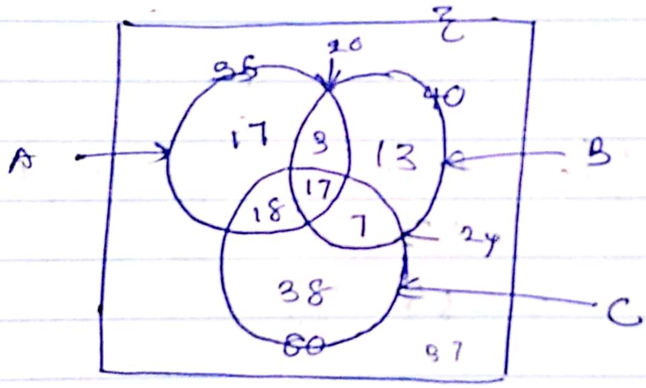




LHS = RHS

$(A-B)\cap C = (A\cap C)-B$

(b)



$(A\cap B)\cap C' = 20-17$
$= 3$

$(B\cap C)\cap A' = 24-17$
$= 7$

$A\cup C = 100$

$A\cup C = C + ((A\cap B)\cap C') + (B\cup C)'\cap A$

$100 = 80 + 3 + (B\cup C)'\cap A.$

$(B\cup C)'\cap A = 100-83 = 17$

$(A\cap C)\cap B' \Rightarrow (A\cap C)\cap B' + 17+3+17 = 55$

$(A\cap C)\cap B' = 55-37 = 18$

(I) $|A\cap C| = 18+17$
$= 35$

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(II) $|(B\cup C) - (A\cap B\cap C)|$

$(A\cup B)'\cap C = 80-(18+17+7)$

$= 38$

$(A\cup C)'\cap B = 40-(17+3+7)$

$= 13$

$(B\cup C) = 80+13+3$

$= 96$

$|(B\cup C) - (A\cap B\cap C)| = |96-17| = |79| = 79$

(III) $\xi = 150$

$|(A\cup B\cup C)'| = 150-(80+17+3+13)$

$= 150-113$

$= 37$

(04) (I) $(A\cup B)-(A\cap B) = (A-B)\cup(B-A)$

* $(A\cup B)-(A\cap B) \subseteq (A-B)\cup(B-A)$

proof : x is arbitary element in $(A\cup B)-(A\cap B)$

$x \in (A\cup B)-(A\cap B)$

$\Rightarrow x \in A\cup B$ and $x \notin A\cap B$

$\Rightarrow x\in A$ or $x\in B$ and $x\notin A\cap B$

Id Says $x \in A-B$ or $x\in B-A$

$\therefore x \in (A-B)\cup(B-A)$

$\therefore (A\cup B)-(A\cap B) \subseteq (A-B)\cup(B-A) \longrightarrow (1)$

* $(A-B)\cup(B-A) \subseteq (A\cup B) - (A\cap B)$

Proof : y is arbitary element in $(A-B)\cup(B-A)$

$y \in (A-B)\cup(B-A)$

$y\in A-B$ or $y\in B-A$

$y \in A \quad y \notin B$. or $y \in B \quad y \notin A$

combining this two condition, we can conclude

1) If y is in A but not in B, then y is in $A \cup B$ but not in $A \cap B$

2) If y is in B but not in A, then y is in $A \cup B$ but not in $A \cap B$

So we can say; $y \in (A \cup B) - (A \cap B)$

i.e. $(A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B) \longrightarrow (2)$

using (1), (2)

$(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

(II) $(A - B) - C = A - (B \cup C) = (A - B) \cap (A - C)$

let $(A - B) - C = A - (B \cup C)$

* $(A - B) - C \subseteq A - (B \cup C)$

x is an arbitary element in $(A - B) - C$

$x \in (A - B) - C$

$x \in A \quad x \notin B$ and $x \notin C$

$x \in A$ and $x \notin B \cup C$

so; $x \in A - (B \cup C)$

$\Rightarrow (A - (B \cup C)) \subseteq A - (B \cup C) \longrightarrow (1)$

* $A - (B \cup C) \subseteq (A - B) - C$

y is an arbitary element in $A - (B \cup C)$

$y \in A - (B \cup C)$

$y \in A$ and $y \notin B \cup C$

$y \in A$ and $y \notin B$ or $y \notin C$

so; $y \in A - B$ or $y \notin C$

so; $y \in (A - B) - C$

$\Rightarrow A - (B \cup C) \subseteq (A - B) - C \longrightarrow (2)$

(1), (2)

$(A - B) - C = A - (B \cup C)$ //

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(III) $A-(B\cap C) = (A-B)\cup(A-C)$

$A-(B\cap C) \subseteq (A-B)\cup(A-C)$

x is an arbitary element in $A-(B\cap C)$

$x \in A-(B\cap C)$

$x\in A$ and $x \notin B\cap C$

$x\in A$ and $x\notin B$ or $x\notin C$

$x\in A$ and $x\notin B$ or $x\in A$ ~~or~~ and $x\notin C$

$x \in (A-B)$ or $x\in (A-C)$

$x\in (A-B)\cup(A-C)$

$\therefore A-(B\cap C)\subseteq (A-B)\cup(A-C) \longrightarrow (1)$

$(A-B)\cup(A-C) \subseteq A-(B\cap C)$

~~x~~ y is an arbitary element in $(A-B)\cup(A-C)$

$y\in (A-B)\cup(A-C)$

$y\in(A-B)$ or $y\in(A-C)$

$y\in A$ and $y\notin B$ or $y\in A$ and $y\notin C$

$\Rightarrow y\in A$ and $y\notin B$ ~~and~~ or $y\notin C$

$\Rightarrow y\in A$ and $y\notin B\cap C$

$\Rightarrow y\in A-(B\cap C)$

i.e. $(A-B)\cup(A-C)\subseteq A-(B\cap C) \longrightarrow (2)$

using (1), (2)

$A-(B\cap C) = (A-B)\cup(A-C)$ //

(5) $A, B, C \subseteq S$

a) I) If $A\cap B=\emptyset$ then $A\subseteq B'$

$A\cap B = \emptyset$ (given)

let x be an arbitary element in A ; $x\in A$

Assume ; $x\in B$

So ; $x\in B$ and it also in A

$x\in B$ and $x\in A$

It says $x\in A\cap B$

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$A-(B\cup C) = (A-B)\cap(A-C) \longrightarrow$ (A)

$A-(B\cup C) \subseteq (A-B)\cap(A-C)$

x is an arbitary element in $A-(B\cup C)$

$x \in A-(B\cup C)$

$x \in A$ and $x \notin B\cup C$

$x \in A$ and $x \notin B$ or $x \notin C$

$x \in A$ and $x \notin B$ and ~~or~~ $x \in A$ and $x \notin C$

$x \in A-B$ and $x \in A-C$

$\therefore$ $x \in (A-B)\cap(A-C)$

$\therefore$ or $A-(B\cup C) \subseteq (A-B)\cap(A-C) \rightarrow$ (1)

y is an arbitary element in $(A-B)\cap(A-C)$

$y \in (A-B)\cap(A-C)$

$y \in (A-B)$ and $y \in (A-C)$

$y \in A$ and $y \notin B$ and $y \in A$ and $y \notin C$.

$y \in A$ and $y \notin B$ or $y \notin C$

$y \in A$ and $y \notin B\cup C$.

$\Rightarrow y \in A-(B\cup C)$

or $(A-B)\cap(A-C) \subseteq A-(B\cup C) \rightarrow$ (2)

using (1), (2)

$A-(B\cup C) = (A-B)\cap(A-C) \longrightarrow$ (B)

using (A), (B)

$(A-B)-C = A-(B\cup C) = (A-B)\cap(A-C)$ //

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But. $A \cap B = \emptyset$ (given)

So our assumption is wrong

$\therefore$ x is not element of B

$\therefore$ $x \in B'$

since $x \in A$ and $x \in B'$

Therefore, $A \subseteq B'$

$\therefore$ $A \cap B \emptyset$ the $A \subseteq B'$ //

II) $A' - B' = B - A$

let x be an arbitary element in $A' - B'$

$x \in A' - B'$

$x \in A'$ and $x \notin B'$; $x \in S$

$\Rightarrow x \in B$ and $x \notin A$

$\Rightarrow x \in B - A$

$A' - B' \subseteq B - A \longrightarrow (1)$

let y be an arbitary element in $B - A$

$y \in B - A$

$y \in B$ and $y \notin A$; $x \in S$

$\Rightarrow y \notin B'$ and $y \in A'$

$\Rightarrow y \in A' - B'$

$B - A \subseteq A' - B' \longrightarrow (2)$

using (1), (2)

$A' - B' = B - A$ //

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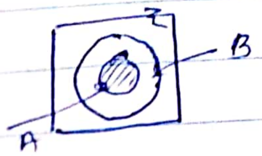

III) $A \subseteq B$

$A \cup (B-A) = B$

let x arbitary element in $A \cup (B-A)$

$x \in A \cup (B-A)$

$\{x \in A$ or $x \in B$ ~~or~~ and $x \notin A\}$ — case1: $x \in B$ — case 2: but $x \in A$ and $x \notin A$

$\cancel{A-B \; (\because A \subseteq B)}$

$\cancel{x \in A \cup (A-A)}$ $\cancel{x \in B \cup (B-B)}$

$\cancel{x \in A \cup \emptyset}$. $\cancel{x \in B \cup \emptyset}$

$\cancel{x \in A}$ $\cancel{x \in B}$

$\Rightarrow$ case 2 is not possible.

$\Rightarrow x \in B$

$\therefore A \cup (B-A) \subseteq B$

$A \cup (B-A) = B$ //

IV) $A \cap (B-C) = (A \cap B) - (A \cap C)$

x be an arbitary element in $A \cap (B-C)$

$x \in A \cap (B-C)$

$x \in A$ and $x \in B-C$

$x \in A$ and $x \in B$ and $x \notin C$

$x \in A$ and $x \in B$ and $x \in A$ and $x \notin C$

$x \in A \cap B$ and $x \in A-C$

$x \in A \cap B$ and $x \notin A \cap C$

so; $x \in (A \cap B) - (A \cap C)$

$A \cap (B-C) \subseteq (A \cap B) - (A \cap C)$

b). $A \Delta (B \Delta C) = (A \Delta B) \Delta C$

x be an arbitary element in $A \Delta (B \Delta C)$

I) $x \in A \Delta (B \Delta C)$

$x \in A \Delta ((B-C) \cup (C-B))$

$\cancel{x \in A - [(B-C) \cup (C-B)] \cup [(B-C) \cup (C-B)] - A}$